

Package ‘GR2MSemiDistr’

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Type Package

Title Semi-Distributed GR2M Model for Large-Sample Hydrological Applications

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Description Implements a semi-distributed adaptation of the GR2M monthly hydrological model for large-sample and operational hydrological modeling. The model uses regional parameters and subbasin-level forcing data to simulate water balance components such as precipitation, evapotranspiration, runoff, and discharge. It supports warm-up periods, initial states, output saving, and regional parameter correction. The routing scheme is based on subbasin structure and spatial relationships derived using TauDEM 5.3.7 (external dependency). Designed for flexible integration in research, water resource assessments, and operational hydrological systems in complex basins.

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Encoding UTF-8

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Create_Forcing_Inputs	<i>Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.</i>
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Description

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

Usage

```
Create_Forcing_Inputs(
  Subbasins,
  Precip,
  PotEvap,
  Qobs = NULL,
  DateIni,
  DateEnd,
  IniGrids = "01/1981",
  Save = FALSE,
  Update = FALSE,
  Members = NULL
)
```

Arguments

Subbasins	A 'SpatVector' object containing the subbasins' geometries. Must include the following attributes: 'COMID' (unique identifier), and 'Region' (string code for the hydro-climatic region).
Precip	A 'SpatRaster' object with monthly precipitation data in [mm/month].
PotEvap	A 'SpatRaster' object with monthly potential evapotranspiration data in [mm/month].
Qobs	Optional. A vector with observed streamflow in [m ³ /s] at the basin outlet.
DateIni	Start date of the output time series in 'mm/yyyy' format.
DateEnd	End date of the output time series in 'mm/yyyy' format.
IniGrids	Initial date of the gridded precipitation and evapotranspiration datasets in 'mm/yyyy' format.
Save	Logical. If TRUE, saves the resulting data frame to a '.txt' file in the './Inputs' directory. Default is FALSE.

Update	Logical. If TRUE, returns only the last month of data (useful for operational updates). Default is FALSE.
Members	Optional. Integer indicating the number of ensemble members for streamflow forecasting. If provided, the date vector will be repeated accordingly. Default is NULL.

Value

A data frame containing the model inputs in the airGR format, with columns:

DatesR Monthly dates

P_x Mean precipitation for subbasin x, where x is the COMID

E_x Mean potential evapotranspiration for subbasin x, where x is the COMID

Q Observed streamflow [m³/s], if provided

References

Aybar C, Fernández C, Huerta A, Lavado W, Vega F, Felipe-Obando O. (2020). Construction of a high-resolution gridded rainfall dataset for Peru from 1981 to the present day. *Hydrological Sciences Journal*, 65(5), 770–785. doi:10.1080/02626667.2019.1649411

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)
data(pisco_pr) # Monthly precipitation SpatRaster
data(pisco_pe) # Monthly potential evapotranspiration SpatRaster
data(qobs)     # Data frame with observed streamflow
data(roi)      # Subbasins SpatVector

# Create model input data
data_inputs <- Create_Forcing_Inputs(
  Subbasins = roi,
  Precip = pisco_pr,
  PotEvap = pisco_pe,
  Qobs = qobs,
  DateIni = '01/1981',
  DateEnd = '12/2016',
  IniGrids = '01/1981'
)
View(data_inputs)
```

Description

Optimizes the GR2M monthly water balance model parameters (X1, X2, fp, fe) for multiple calibration regions using the SCE-UA algorithm. The model is run in a semi-distributed setup over multiple subbasins grouped by region. Users can select from various objective functions (e.g., NSE, KGE, RMSE, OF1–OF10), including composite metrics. Parameter bounds must be provided, and regions can be excluded from calibration.

Usage

```
Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = 1500,
  Optimization = "OF1",
  WarmUp = 36,
  No.Optim = NULL,
  w1 = 0.6,
  w2 = 0.3,
  w3 = 0.2
)
```

Arguments

Data	A data frame of model input data in the airGR format, as produced by Create_Forcing_Inputs. It must include columns: DatesR, P_1 to P_n, E_1 to E_n, and Q (used for calibration).
Subbasins	A SpatVector object containing the geometries of subbasins. It must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).
RunIni	Simulation start date in the format "mm/yyyy".
RunEnd	Simulation end date in the format "mm/yyyy".
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.
Parameters.Min	Numeric vector of length 4, specifying lower bounds for optimization in the following order: c(X1, X2, fp, fe).
Parameters.Max	Numeric vector of length 4, specifying upper bounds for optimization in the same order as 'Parameters.Min'.
Optimization	Character. Objective function to optimize. Available options include: "OF1" Kling-Gupta Efficiency (KGE). "OF2" Nash-Sutcliffe Efficiency (NSE). "OF3" NSE applied to log-transformed flows (log-NSE, Pushpalatha 2012). "OF4" Root Mean Square Error (RMSE). "OF5" Percent Bias (PBIAS). "OF6" FDC Bias (FDC Bias), based on flow duration curves. "OF7" Pearson correlation coefficient (r). "OF8" Composite objective: $w1 \times KGE + w2 \times NSE - w3 \times RMSE$.

	"OF9" Composite objective: $w1 \times KGE + w2 \times \log\text{-NSE} - w3 \times \text{RMSE}$.
	"OF10" Composite objective: $w1 \times KGE.\text{km} + w2 \times KGE.\text{lf} - w3 \times \text{RMSE}$.
No.Optim	Character vector (optional). Region names to exclude from the optimization (parameters will be kept fixed).
w1	Numeric. Weight for the first component in composite objective functions (OF8, OF9, OF10). Default is 0.6.
w2	Numeric. Weight for the second component in composite objective functions. Default is 0.3.
w3	Numeric. Weight for the third component in composite objective functions. Default is 0.2.
Max.Functions	Integer. Maximum number of function evaluations in the optimization. Default is 1500.

Value

A list with the following elements:

Parameters Data frame of optimized parameters per region (columns: Region, X1, X2, fp, fe).

OF Final value of the selected objective function.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
# Optimize parameters for two regions with KGE as objective function
Parameters <- data.frame(Region = c("A", "B"),
  X1 = c(500, 600),
  X2 = c(1, 1),
  fp = c(1, 1),
  fe = c(1, 1))

result <- Optim_GR2MSemiDistr(Data = data,
  Subbasins = shp,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  WarmUp = 12,
  Parameters = Parameters,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = c(2000, 10, 1.2, 1.2),
  Max.Functions = 1000,
  Optimization = "KGE")

best_params <- result$Parameters
final_score <- result$OF
```

Routing_GR2MSemiDistr *Routing discharges for each subbasin.*

Description

Routing discharges for each subbasin.

Usage

```
Routing_GR2MSemiDistr(
  Model,
  Subbasins,
  Dem,
  AcumIni = NULL,
  AcumEnd = NULL,
  Save = FALSE,
  Update = FALSE
)
```

Arguments

Model	List of model results from Run_GR2MSemiDistr.
Subbasins	Subbasins' shapefile. Must contain the following attributes: 'Area' (in km2), 'Region' (in letters), and 'COMID' (identifier number).
Dem	Digital elevation model raster for the extent of the basin.
AcumIni	Initial date of the model routing in 'mm/yyyy' format. NULL as default
AcumEnd	Ending date of the model routing in 'mm/yyyy' format. NULL as default
Save	Boolean to save results as a text file in the 'Outputs' location. FALSE as default.
Update	Boolean for the updating mode where only the last month's values will be returned. FALSE as default.

Value

List of model routing outputs.

QR: Routed discharge timeseries for all subbasins in [m3/s].

Dates: Vector of dates of the simulation period.

COMID: Vector of identifier numbers for each subbasin.

Author(s)

Llaucha H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*. 2021; 13(8):1048. <https://doi.org/10.3390/w13081048>

Examples

```
# Load data
library(GR2MSemiDistr)
data(dem)

# Routing discharges in the streamflow network
rou <- Routing_GR2MSemiDistr(Model=model,
                             Subbasins=roi,
                             Dem=dem,
                             AcumIni='01/1981',
                             AcumEnd='12/2016')

View(rou$QR)
```

Run_GR2MSemiDistr	<i>Run the GR2M model for multiple subbasins</i>
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Description

This function executes the GR2M monthly water balance model across multiple subbasins using a semi-distributed approach. It applies region-specific parameters and correction factors for precipitation and evapotranspiration, and optionally saves the results as text files.

Usage

```
Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  WarmUp = NULL,
  IniState = NULL,
  Save = FALSE,
  Update = FALSE
)
```

Arguments

Data	A data frame of model input data in the airGR format, as produced by <code>Create_Forcing_Inputs</code> . It must include columns: <code>DatesR</code> , <code>P_1</code> to <code>P_n</code> , <code>E_1</code> to <code>E_n</code> , and optionally <code>Q</code> .
Subbasins	A <code>SpatVector</code> object containing the geometries of subbasins. It must include attributes <code>"COMID"</code> (unique subbasin ID) and <code>"Region"</code> (region name/code).
RunIni	Simulation start date in the format <code>"mm/yyyy"</code> .
RunEnd	Simulation end date in the format <code>"mm/yyyy"</code> .
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: <code>Region</code> , <code>X1</code> , <code>X2</code> , <code>fp</code> , <code>fe</code> .
WarmUp	Optional number of months to discard from the beginning of the simulation as warm-up. Default is <code>NULL</code> .
IniState	Optional list of initial states for each subbasin. Default is <code>NULL</code> .

Save	Logical. If TRUE, output time series will be saved as text files in the "Outputs/" folder.
Update	Logical. If TRUE and Save = TRUE, it will remove previous month's output files before saving the current ones.

Value

A list with the following elements:

PR Matrix of total precipitation [mm/month] for each subbasin

AE Matrix of actual evapotranspiration [mm/month] for each subbasin

SM Matrix of production store level [mm/month] for each subbasin

SM Matrix of percolation [mm/month] for each subbasin

RU Matrix of runoff [mm/month] for each subbasin

QS Matrix of flow [m³/s] (not routed) for each subbasin

Dates Date vector for the simulation period

COMID Vector of subbasin COMIDs

EndState List of final state variables for each subbasin

SINK Optional. Data frame of simulated and observed outlet discharge, if observed Q is provided

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
# Run the GR2M model
model <- Run_GR2MSemiDistr(
  Data = data,
  Subbasins = roi,
  RunIni = "01/1981",
  RunEnd = "12/2016",
  Parameters = data.frame(
    Region = c("A"),
    X1 = 10.5,
    X2 = 0.8,
    fp = 1.1,
    fe = 1.0
  )
)
```


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